Learning Latent Representations with Progressive Hypothesis Space Expansion

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Abstraction in Phonology

- Debate in phonology: **How abstract can Underlying Representations (URs) be?** (Kenstowicz and Kisseberth, 1977; Kiparsky, 1973)

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/tɑ/

concrete URs

less abstract ⟷ *more abstract*

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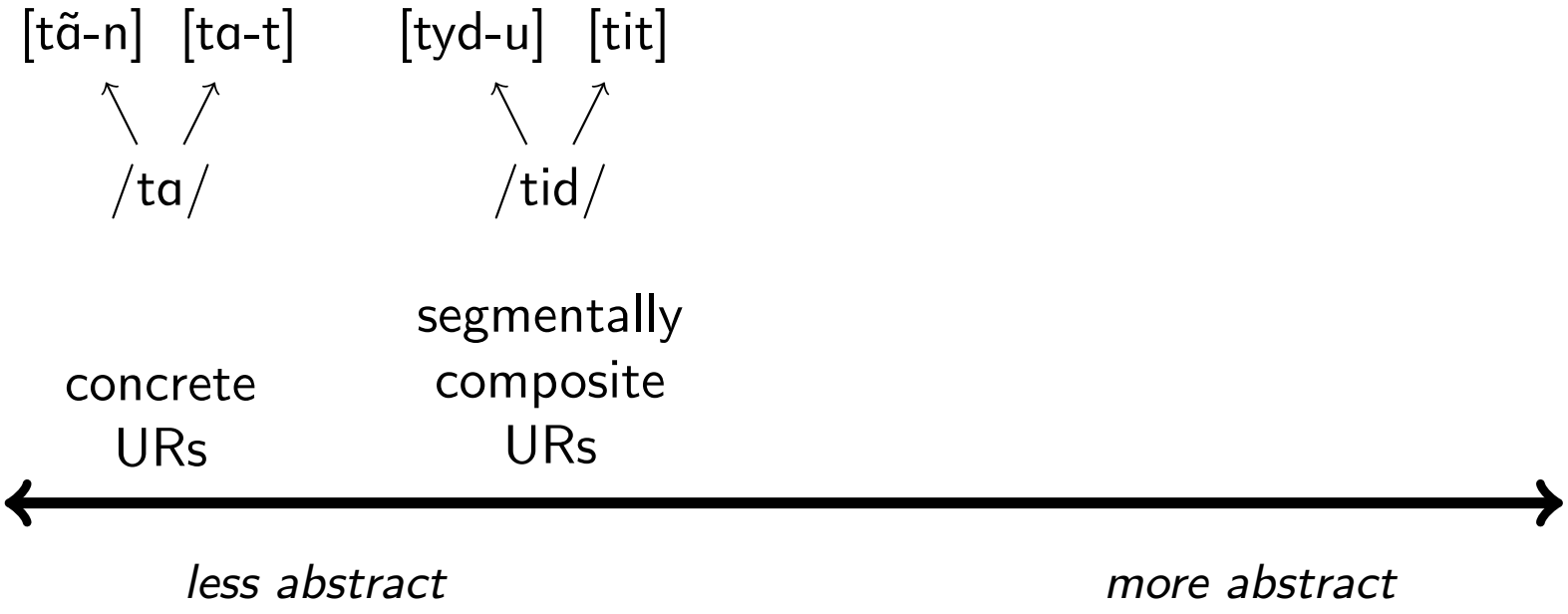

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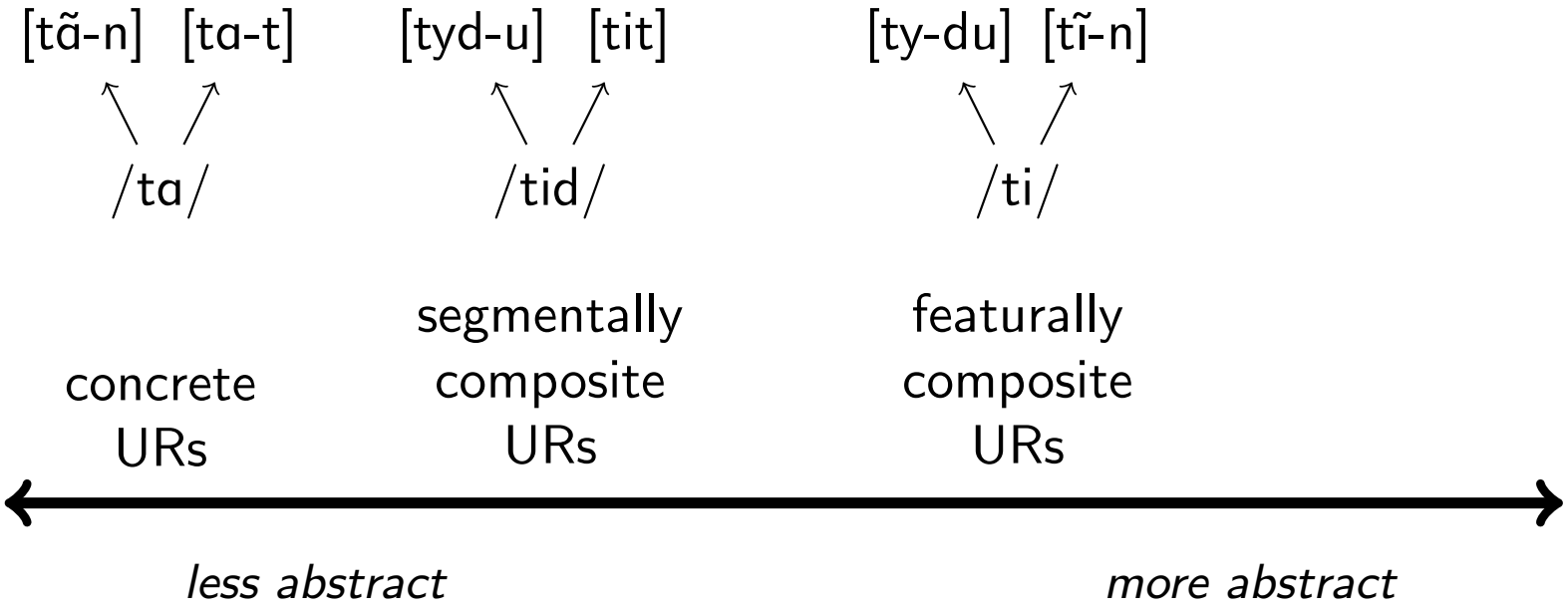

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[tã-n] [tɑ-t]	[tyd-u] [tit]	[ty-du] [tĩ-n]	[tãn-ə] [tãn]
/tɑ/	/tid/	/ti/	/tɑn/
concrete URs	segmentally composite URs	featurally composite URs	covert URs

less abstract ⟷ *more abstract*

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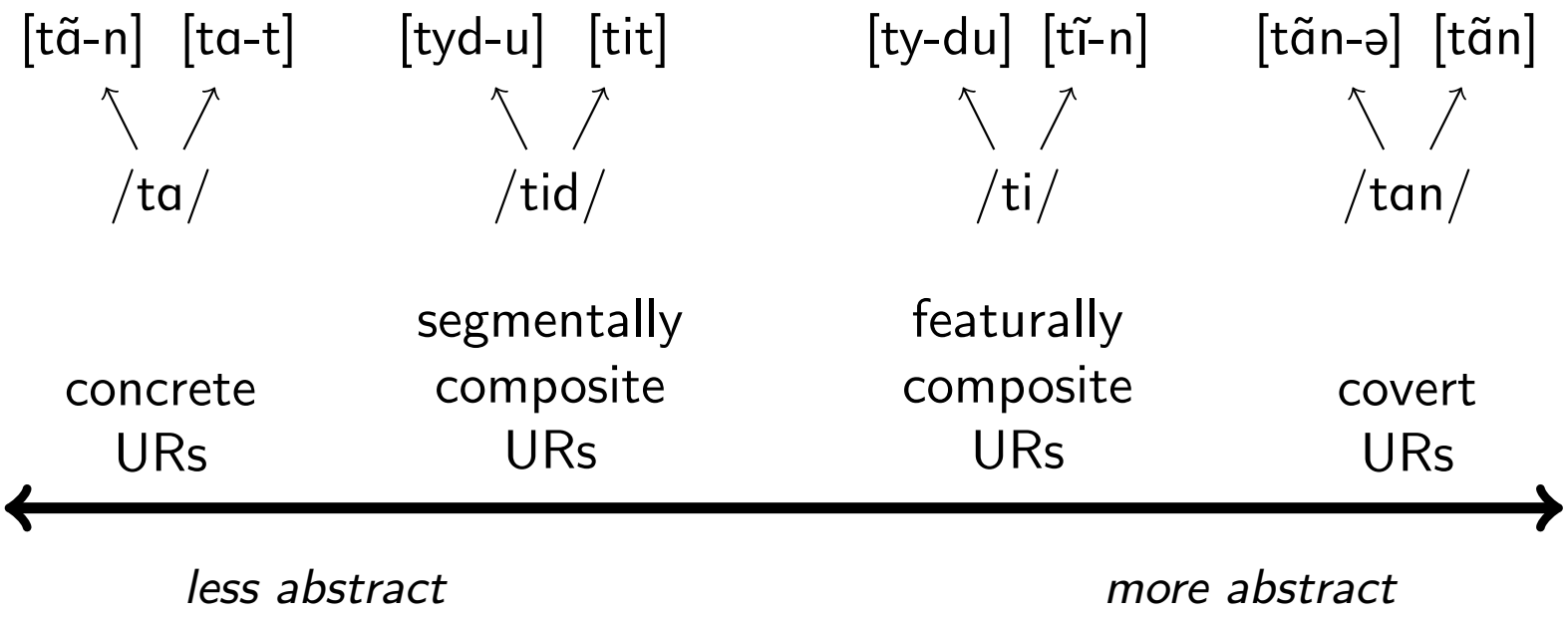

- Covert URs are analytically necessary (e.g., Punjabi vowel nasality)

Vowel Nasality in Punjabi

- Vowel nasality is contrastive (Bashir and Conners, 2019)

[sɑ̃ː] 'I was' vs. [sɑː] 'breath'

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  [sɑ̃ːɳ] 'grindstone' vs. *[sɑːɳ]

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  [sɑ̃ːɳ] 'grindstone' vs. *[sɑːɳ]

- pre-N vowels are phonetically identical to contrastive nasal vowels in terms of nasality (Paramore and Bennett, 2025)

Nasal Harmony in Punjabi

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- Contrastive nasal vowels trigger nasal harmony (Bhatia, 1993)

/sɑː-ʊãː/	→	[sãːʊ̃ãː]	'breath-PL'
/mətʃiː-jãː/	→	[mətʃĩːj̃ãː]	'fish-PL'

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- pre-N vowels **do not** trigger nasal harmony (Paramore and Bennett, 2025)

/ɑːʊɑːm/	→	[ɑːʊãːm]	'public'
/siːjɑːɳ/	→	[siːjãːɳ]	'recognition'

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- Simplest analysis treats pre-N vowels as **covert**: they are underlyingly oral but always surface as nasal
  - only underlying nasal vowels trigger harmony

Abstraction & Learnability

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Abstraction raises two general learnability problems (Jarosz, 2019):

1. **Learners should prefer less abstract URs whenever possible.**
   - How is this preference enforced?

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   - Computationally infeasible to search an infinite space for the optimal UR

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2. **Permitting covert URs makes the hypothesis space computationally intractable.**
   - Computationally infeasible to search an infinite space for the optimal UR

Covert URs should not be permitted because they result in learning failures (Wang and Hayes, 2025)

UR Progressive Hypothesis Space Expansion Learner

UR PHaSE Learner solves both learnability issues by searching through the hypothesis space progressively

- Considers URs one disparity level at a time
- Stops as soon as it finds a satisfactory explanation of the data

(Simon, 1955, 1956)

UR PHaSE Learner

Four iterative learning steps

Step I: phonotactic learning

UR PHaSE Learner

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Step II: UR Expectation Maximization learning

- Determines optimal grammar
- Using a finite set of UR candidates

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- Evaluates whether observed data is sufficiently likely under the current grammar
- Learning successful if likelihood of SRs $> .95$

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Case Study: Punjabi Vowel Nasality

Input to the Learner

- Small Punjabi lexicon (30 forms) that demonstrates the nasality patterns, e.g.,:

i.	[sɑː]	'breath'	ii.	[sɑ̃ː-ʊ̃ɑ̃ː]	'breath-PL'
iii.	[gɑ̃ː]	'cow'	iv.	[gɑ̃ː-ʊ̃ɑ̃ː]	'cow-PL'
v.	[tɑːʊɑ̃ːɳ]	'penalty'			

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v.	**[tɑːʊɑ̃ːɳ]**	**'penalty'**			

- 10 constraints (see appendix for definitions)
  - 9 standard faithfulness/markedness constraints
  - 1 contextual faithfulness constraint (Hauser and Hughto, 2020) to handle opaque application of nasal harmony

Step I: Phonotactic Learning

Goal: Calibrate constraint weights to acquire phonotactic patterns

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Constraint	Type	initial w	final w
SPRD-L[NAS]	mark.	50.00	0.13
*NASOBS	mark.	50.00	100.00
*NASG	mark.	50.00	0.47
*NASV	mark.	50.00	0.00
*VN	mark.	50.00	100.00
*LOWRD	mark.	50.00	100.00

Table 1: Markedness constraint weights during phonotactic learning

Step II: EM Learning with 0-disparity URs

Assumption of the learner: Introduce UR→SR disparities (abstraction) only when learning fails at lower disparity levels.

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Morpheme	'breath'		'penalty'
SRs	[sɑː]	[sɑ̃ː]	[tɑːʊɑ̃ːɳ]
0-disparity URs			

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0-disparity URs	sɑː	sɑ̃ː	

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SRs	[sɑː]	[sɑ̃ː]	[tɑːʊɑ̃ːŋ]
0-disparity URs	sɑː	sɑ̃ː	tɑːʊɑ̃ːŋ

Step II: EM Learning with 0-disparity URs

- Use EM learning to search for the combination of URs and weights that maximize data likelihood (Jarosz, 2006a,b, 2009, 2010, 2015)

Morpheme	UR candidates	Prior $\mathbb{P}$
'breath'	/sɑː/	0.5
	/sɑ̃ː/	0.5
'penalty'	/tɑːʊɑ̃ːŋ/	1.0

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Constraint	initial w	final w
SPRD-L[NAS]	0.13	62.92
*NASOBS	100.00	100.00
*NASG	0.47	60.24
*NASV	0.00	23.16
*VN	100.00	100.00
ID[NAS]	0.00	83.44
IDFin[NAS]	0.00	89.46
ID[NAS]/_V	0.00	100.00
ID[RD]	0.00	6.29
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'penalty'	/tɑːʋɑ̃ːɳ/	1.0	1.0

- No satisfactory solution exists in the 0-disparity hypothesis space
  - Learner can't jointly model harmony in [sɑ̃ːʋ̃ɑ̃ː] and non-harmony in [tɑːʋɑ̃ːɳ]

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	/sɑ̃ː/	0.5	**0.5**
'penalty'	/tɑːʋɑ̃ːɳ/	1.0	1.0

- No satisfactory solution exists in the 0-disparity hypothesis space
  - Learner can't jointly model harmony in [sɑ̃ːʋ̃ɑ̃ː] and non-harmony in [tɑːʋɑ̃ːɳ]
  - Distributes probability across competing URs for 'breath'

Step III: Likelihood Threshold Evaluation

SR	Likelihood
[sɑː]	0.50
[sɑ̃ːʊ̃ɑ̃ː]	0.48
[tɑːʊɑ̃ːɳ]	1.00

Likelihood Threshold of $> .95$ not met!

Step IV: UR Candidate Expansion

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  - Learner considers more abstract URs

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  - Identify markedness constraints at end of EM learning with intermediate weights, violated by 1+ SR

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  - Learner considers more abstract URs
- Expand the UR candidate space of each morpheme by introducing 1-disparity URs
- Expansion does not include all features
  - Identify markedness constraints at end of EM learning with intermediate weights, violated by 1+ SR
  - Only features referenced by those constraints are eligible to be flipped to create new URs

Step IV: UR Candidate Expansion

- Only the [nas] feature is eligible to be flipped

Constraint	final w
SPRD-L[NAS]	62.92
*NASOBS	100.00
*NASG	60.24
*NASV	23.16
*VN	100.00
*LOWRD	100.00

Step IV: UR Candidate Expansion

- Expand UR space: create 1-disparity candidates by flipping [NAS] on eligible segments

Morpheme	'breath'		'penalty'
SRs	[sɑː]	[sɑ̃ː]	[tɑːʊɑ̃ːɳ]
0-disparity URs	sɑː	sɑ̃ː	tɑːʊɑ̃ːɳ

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Morpheme	'breath'		'penalty'
SRs	[sɑː]	[sɑ̃ː]	[tɑːʊɑ̃ːɳ]
0-disparity URs	sɑː	sɑ̃ː	tɑːʊɑ̃ːɳ
1-disparity URs			

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Morpheme	'breath'		'penalty'
SRs	[sɑː]	[sɑ̃ː]	[tɑːʊɑ̃ːŋ]
0-disparity URs	sɑː	sɑ̃ː	tɑːʊɑ̃ːŋ
1-disparity URs		tɑːʊ̃ɑ̃ːŋ	

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Morpheme	'breath'		'penalty'
SRs	[saː]	[sãː]	[taːʋãːŋ]
0-disparity URs	saː	sãː	taːʋãːŋ
1-disparity URs		taːʋ̃ãːŋ	tãːʋãːŋ

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SRs	[saː]	[sãː]	[taːʋãːŋ]
0-disparity URs	saː	sãː	taːʋãːŋ
1-disparity URs		taːṽãːŋ	tãːʋãːŋ, taːʋaːŋ

Step II: EM Learning with 1-disparity URs

Constraint	initial w	final w
SPRD-L[NAS]	0.13	48.45
*NASOBS	100.00	100.00
*NASG	0.47	48.45
*NASV	0.00	0.00
*VN	100.00	100.00
ID[NAS]	0.00	0.00
IDFin[NAS]	0.00	100.00
ID[NAS]/_V	0.00	100.00
ID[RD]	0.00	6.28
*LOWRD	100.00	100.00

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Constraint	initial w	final w
SPRD-L[NAS]	0.13	48.45
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ID[RD]	0.00	6.28
*LOWRD	100.00	100.00

Morpheme	UR candidates	Prior $\mathbb{P}$	Posterior $\mathbb{P}$
'breath'	/saː/	0.5	**1.0**
	/sãː/	0.5	0.0
'penalty'	/taːʋãːɳ/	0.25	0.0
	/taːʋaːɳ/	0.25	**1.0**
	/taːʋ̃ãːɳ/	0.25	0.0
	/tãːʋãːɳ/	0.25	0.0

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	/sãː/	0.5	0.0
'penalty'	/taːʋãːɳ/	0.25	0.0
	/taːʋaːɳ/	0.25	**1.0**
	/taːʋ̃ãːɳ/	0.25	0.0
	/tãːʋãːɳ/	0.25	0.0

- 1-disparity UR space resolves the learning problem
  - Successfully models harmony in [sãːʋ̃ãː]
  - Successfully models non-harmony in [taːʋãːɳ]

Step III: Likelihood Evaluation

SR	Likelihood
[saː]	1.00
[sãːṽãː]	0.96
[taːʋãːɳ]	1.00

Likelihood Threshold satisfied!

A Solution to Learnability Problems

1. **Preference for minimally abstract representations**
   - Abstraction is introduced only to the extent necessary by nature of the serial expansion of UR candidate space
2. **Search Space Tractibility**
   - An unbounded search space can be searched efficiently if:
     (i) the space is structured appropriately
     (ii) search for a satisfactory solution rather than the optimal one

Thank you!

References I

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Appendix: Constraint Definitions I

i. SPRD-L[NAS] (cf. Walker, 2003, p.47)
For every occurrence of a [+NAS] feature in a prosodic word, if that [+NAS] feature is dominated by some segment, assign a violation for every segment to the left of that segment that does not dominate the [+NAS] feature.

ii. *NASOBS (Walker, 2003, p.51)
Assign a violation for every obstruent that dominates a [+NAS] feature.

iii. *NASG (Walker, 2003, p.51))
Assign a violation for every glide that dominates a [+NAS] feature.

iv. *NASV (Walker, 2003, p.51))
Assign a violation for every vowel that dominates a [+NAS] feature.

v. *VN
Assign a violation for every vowel that dominates a [-NAS] feature when directly preceding a nasal consonant.

vi. ID[NAS]
For every segment, A, assign a violation if the output value for the [NAS] feature dominated by A does not match the input value for the [NAS] feature dominated by A.

Appendix: Constraint Definitions II

vii. IDFIN[NAS]
For every segment, *A*, assign a violation if the output value for the [NAS] feature dominated by *A* does not match the input value for the [NAS] feature dominated by *A* in the final syllable of a prosodic word.

viii. ID[NAS]/__V
Let *A* be a segment that occurs before an oral vowel, V, in the input. Assign one violation if the output correspondent of *A* does not have the same specifications for [NAS] as *A*.

ix. ID[RD]
For every segment, *A*, assign a violation if the output value for the [RD] feature dominated by *A* does not match the input value for the [RD] feature dominated by *A*.

x. *LOWRD
Assign a violation for every vowel that dominates a [RD] feature and a [LOW] feature simultaneously.